

Elizaveta Rebrova

Math PhD Student / University of Michigan

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Education

University of Michigan / PhD in Mathematics

2013 - May 2018(expected), Ann Arbor, MI

Thesis title "Spectral properties of heavy-tailed random matrices"; advisor Roman Vershynin

Research area: random matrix theory; high-dimensional probability and geometry; random graphs and other applications to high-dimensional data analysis

Lomonosov Moscow State University / Specialist in Mathematics (=B.S.+M.S.)

2007 - 2012, Moscow, Russia

Diploma with honors, 5.00/5.00 GPA

Full undergraduate math major coursework

Advanced level coursework: Real and Complex Analysis (specialization), Probability, Statistics, Control and Optimization, Algebra, Number Theory, Discrete Mathematics

Master's thesis "Sobolev classes on infinite dimensional spaces"; Advisor V.I. Bogachev

Yandex Data Analysis School / Two year on-site program on Computer Science

2010 - 2012, Moscow, Russia

12 Masters-level courses, including Algorithms and Data Structures with C++, Complexity Theory, Machine Learning, Computational Linguistics, Game Theory, Intro R, Applied Statistics

Experience

Graduate internship at Lawrence Berkeley National Lab

July-August 2017, Berkeley, CA

Scalable Solvers Group; internship mentor: Xiaoye Sherry Li

Applied STRUMPACK linear solver <http://portal.nersc.gov/project/sparse/strumpack/> to kernel matrices, resulting in a new approach to perform training phase of supervised ML algorithms with the kernel trick fast and in parallel, tested on kernel ridge regression

Employed several clustering algorithms to preprocess matrices for the STRUMPACK algorithm, resulting in up to 10 times better compression comparing to the naive application

Experiments and implementation were performed in Matlab and C++

Now working on a conference paper based on the results obtained

Research project at Universite Paris-Est Marne-la-Vallee

June 2017, Champs-sur-Marne, France

Investigated theoretical requirements for randomized dimension reduction. Collaborators: Professors Olivier Guedon and Pierre Youssef

Teaching and service

over 10 years, Russia and US

2013 - 2017 Graduate Student Instructor at UofMichigan

- one-to-one tutoring for all junior and sophomore Math courses
- primary instructor for Calculus: prepared and conducted lectures, organized and supervised team work, created and graded weekly quizzes
- teaching assistant for Differential Equations (led recitations and Matlab labs) and Computational Finance classes

2009 - 2013 Seminar work with high school students, Moscow State School N 57 and Kolmogorov Math and Physics high school

2007 - 2013 Member of organizing committee, Moscow Mathematical and Linguistics Olympiads for high school students

Publications

2016 Norms of random matrices: local and global problems (with R. Vershynin), accepted to Advances in Mathematics

2016 Coverings of random ellipsoids, and invertibility of matrices with i.i.d. heavy-tailed entries (with K. Tikhomirov), accepted to Israel Journal of Mathematics

2013 Functions of bounded variation on infinite-dimensional spaces with measures (with V. I. Bogachev). Doklady Mathematics, 87 (2)

2013 Classes of functions of bounded variation on infinite-dimensional domains. (with V. I. Bogachev and A.Y. Pilipenko). Doklady Mathematics, 88 (1)

Presentations (selected)

September 2017 "STRUMPACK meets kernel matrices", summer project presentation for the research team, LBNL, Berkeley, CA

March 2017 "Regularization of the random matrix norm: local and global obstructions", research seminar talk at Center for Data Science, NYU, New York

June 2016 "Regularization of the norm of matrices with heavy-tailed entries", Canadian Math Society Summer Meeting talk, Alberta, Canada

March, 2016 "Covering of random ellipsoids and Invertibility of matrices with heavy-tailed entries", American Math Society Sectional Meeting talk, Athens, GA

Skills

Mathematics

Main specialization for over 15 years

Concentration in Probability and Geometry of High-Dimensional Spaces

PhD research experience, independent and with collaborators from US, Canada and France

Programming

Python (96.27 percentile in hackerrank online contests with about 15 contests over the last 2 years + data wrangling with python)

Matlab (wrote around 10 machine learning related methods for linear solver STRUMPACK during LBNL 2017 summer internship + teaching DiffEq modeling labs at UofMichigan)

C++ (coursework; improving existing STRUMPACK code to extend functionality)

Machine learning

Used data analytics tools, including jupyter, pandas, numpy, scikit-learn. Recent project: performed analytics and wait time prediction for the jobs submitted on NERSC cluster for LBNL internal data competition project

Participated in the Yandex Data School team projects, including top-3 in class in Authorship Attribution project (automatic detection of Russian poems authors) and top-1 in Italian Morphology project (automatic detection of lemma and part of speech of Italian words)

Led about 16 sessions of machine learning seminar during LBNL 2017 summer internship

Languages

English - fluent; Russian - native; French and German - basic

Grants and awards

Spring 2018 Allen Shields Memorial Fellowship

Summer 2017 NSF internship at Lawrence Berkeley National Lab

Spring 2017 Rackham Dissertation writing fellowship, UofMichigan

2010 - 2012 Excellent progress mayor's scholarship and research grant RFFI 10-01-00518, Moscow, Russia

Spring 2007 No-exam entry to Moscow State University Math department due to the award place at Moscow Math Olympiad

July 2007 The 5th International Linguistics Olympiad (for high school students), 1st place team award and individual Bronze award